Capstone Project - API Tester

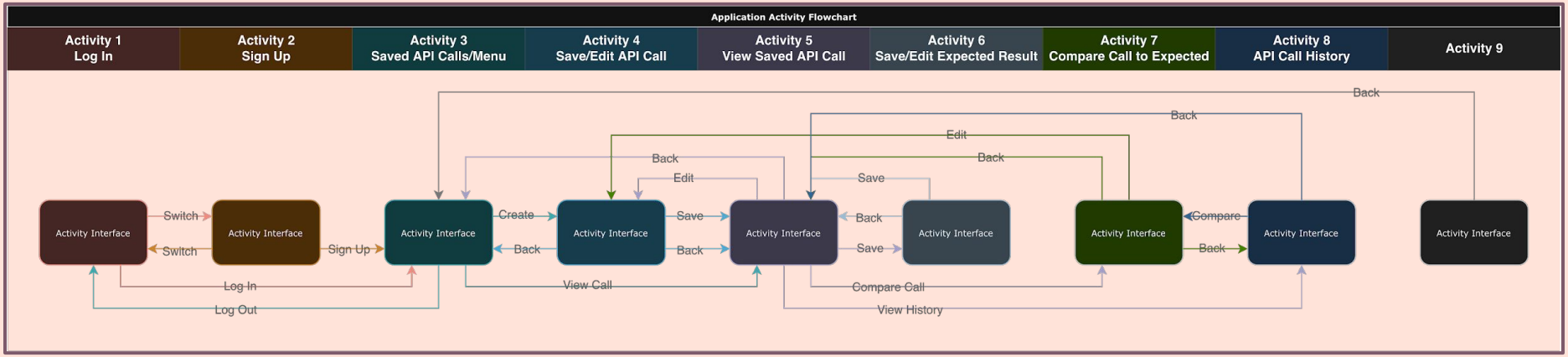
Group 12 - Jason Rabinowitz and Eric Vu

Goal:
save, view, and compare json responses from user issued API calls

App Activities

Main user features and interactions of the application

1. Login Activity
2. Signup Activity
3. Saved API Calls Menu Activity
4. Save/Edit API Calls Activity
5. View Saved API Call Activity
6. Save/Edit Expected JSON Response Activity
7. Compare API Call to Expected Response Activity
8. API Call History Activity
9. About the App Activity

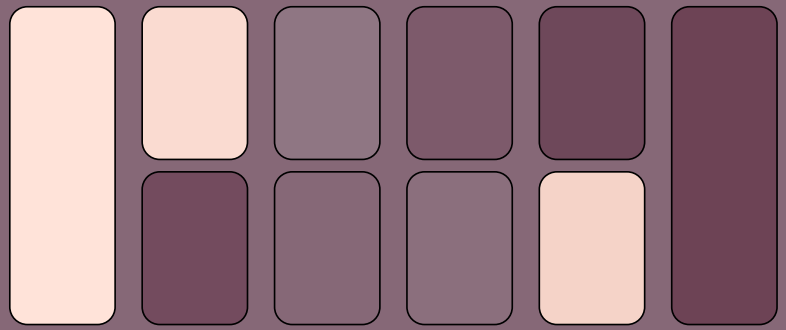


Project Activity Flowchart

UI Color Palette

694455
6E485A
734B5E
7D5A6B
866877
8B6F7D
8F7683
F5D3C8
FADBD1
FFE3D9

Color Interactions



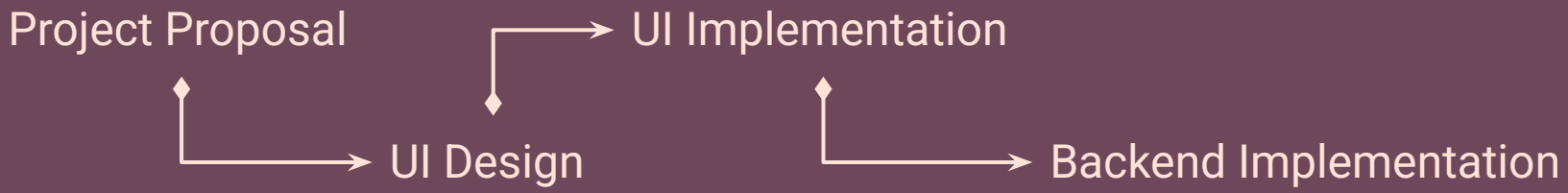
Frontend Implementation Description

The frontend is a unified design that uses xml style sheets in order to drive a consistent design across all activities. Elements also include drawable backgrounds with slightly rounded corners for aesthetics. The UI is designed to fit the vertical orientation of the device in order to keep consistent spacing and formatting on rotation. In addition to recyclerviews for element scrolling, the application utilizes scrollviews so that all elements are able to be accessed by the user. Each of the back buttons on the activities are set to only go back to the previous activity via the finish() method rather than starting a new activity via intents in order to keep a consistent application flow.

Backend Implementation Description

The backend of this app utilizes Firebase Authentication to manage user accounts, as well as using Cloud Database to save API calls. Firebase Authentication handles user sign-up, login, and logout using email and password credentials, and once authenticated, each user is uniquely identified by their uid, which is used to scope all database operations. Cloud Firestore is used to store saved API call data under user-specific collections, ensuring that each user can only access their own saved requests. Saved APIs store the API URL, expected JSON, actual response JSON, date and time, and API key into the database for references. The Database also starts a collection of history of tests and details about each test and gives the date and time of each one.

Project Timeline



Group Member Contributions

Jason Rabinowitz

Frontend design and implementation

Eric Vu

Backend design and implementation